SWE 600 ADVANCED PARALLEL ALGORITH DESIGN AND ANALYSIS
PROJECT STUDY 1 (2026)

Due date: 28 Oct 2026, in class.

In this project, your task is to design, implement and evaluate a parallel image processing software (convolution algorithm, CNN) using: 1) Pthreads, 2) OpenMP and 3) Multi-process model. Initially you are expected to implement a serial model and then re-design the algorithm for optimum parallel performance. For each implementation, you are expected to measure the performance (execution time) and calculate the speedup against the serial run. You can use either C or C++ language for your implementations. The software is supposed to do image filtering on BMP images by convolution and ReLU as shown in Figure 1. Input image files are supposed to be BMP format and 4K size.

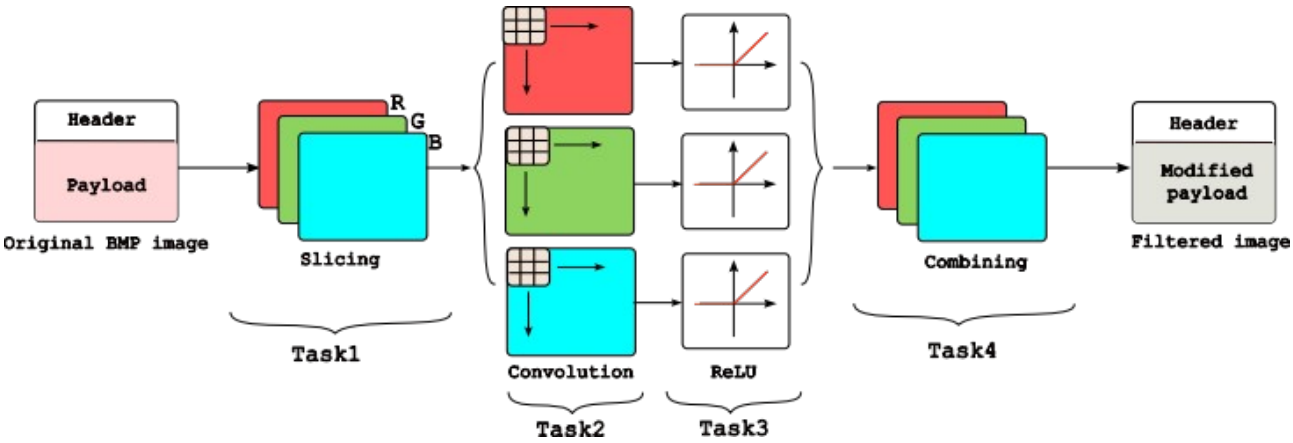
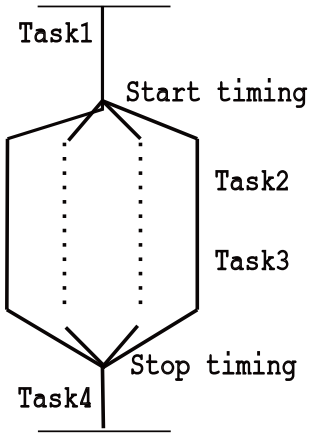


Figure 1. Project block diagram. Tasks show different stages of computation.

0	-1	0
-1	5	-1
0	-1	0

(a)



(b)

Figure 2. a) Example CNN kernel, b) Program activity graph (PAG).

An example 3X3 kernel is shown in Figure 2.a., which you can use in your implementation. However, it is expected that your implementation allow switching other kernels such as sharpening, blurring etc. without re-compilation. Figure 2.b. shows how tasks are going to be executed in time.

Do the following experiments to complete you project work:

- Implement and measure timing for serial implementation. Timestamp your application same location as shown in Figure 2.b.
- Implement Pthread version with 3, 6 and 9 threads, measure execution time as shown in Figure 2.b. and calculate the speedup.
- Implement OpenMP version with 3, 6 and 9 threads, measure execution time as shown in Figure 2.b. and calculate the speedup.

- Implement multi process solution that will run on single comuter for 3, 6 and 9 processes. Use scoket interface for inter-process communication. If you wish, you can use shared memory approach as well. Measure execution time as shown in Figure 2.b. and calculate the speedup.
- This is optional: For each model repeat the experiments (and time measuring) at least 5 times and take average execution time to reduce cache effects.
- Put your results in a graph; X axis will show thread/process count and Y axis will show speedup.
- Comment/discuss your findings in the presentation.

No	Task	Grade
1	Serial and parallel programming models work without error.	**60**
2	Performance evaluations are done correctly and results are meaningful.	**20**
3	Presentation and discussions are OK.	**20**

PS: In class presentation is required.